Neuroscience 248 (2013) 552–561

PROLACTIN FRACTIONS FROM LACTATING RATS ELICIT EFFECTS UPON SENSORY SPINAL CORD CELLS OF MALE RATS

F. MENA,[a†] A. GONZÁLEZ-HERNÁNDEZ,[b] N. NAVARRO,[a] A. CASTILLA,[a] T. MORALES,[a] G. ROJAS-PILONI,[b] G. MARTÍNEZ-LORENZANA[b] AND M. CONDÉS-LARA[b*]

[a] Departamento de Neurobiología Celular y Molecular, Instituto de Neurobiología, Universidad Nacional Autónoma de México, Campus UNAM-Juriquilla 3001, Querétaro 76230, Mexico

[b] Departamento de Neurobiología del Desarrollo y Neurofisiología, Instituto de Neurobiología, Universidad Nacional Autónoma de México, Campus UNAM-Juriquilla 3001, Querétaro 76230, Mexico

Abstract—Recently it has been suggested that the neurohormone prolactin (PRL) could act on the afferent nociceptive neurons. Indeed, PRL sensitizes transient receptor potential vanilloid 1 (TRPV1) channels present in nociceptive C-fibers and consequently reduces the pain threshold in a model of inflammatory pain. Accordingly, high plasma PRL levels in non-lactating females have been associated with several painful conditions (e.g. migraine). Paradoxically, an increase of PRL secretion during lactation induced a reduction in pain sensitivity. This difference could be attributed to the fact that PRL secreted from the adenopituitary (AP) is transformed into several molecular variants by the suckling stimulation. In order to test this hypothesis, the present study set out to investigate whether PRL from AP of suckled (S) or non-suckled (NS) lactating rats affects the activity of the male Wistar wide dynamic range (WDR) neurons. The WDR neurons are located in the dorsal horn of the spinal cord and receive input from the first-order neurons (Ab-, Ad- and C-fibers). Spinal administration of prolactin variant from NS rats (NS-PRL) or prolactin variant from S rats (S-PRL) had no effect on the neuronal activity of non-nociceptive Ab-fibers. However, the activities of nociceptive Ad-fibers and C-fibers were: (i) increased by NS-PRL and (ii) diminished by S-PRL. Either NS-PRL or S-PRL enhanced the post-discharge activity. Taken together, these results suggest that PRL from S or NS lactating rats could either facilitate or depress the nociceptive responses of spinal dorsal horn cells, depending on the physiological state of the rats.

Key words: lactation, prolactin, nociception, WDR neurons, rats.

INTRODUCTION

Since its discovery, considerable literature has built up describing more than 300 biological actions of prolactin (PRL) Ben-Jonathan et al., 1996; Bole-Feysot et al., 1998; Grattan and Kokay, 2008). Recent behavioral studies have shown that PRL generated during peripheral inflammation could act as a key mediator to reduce the pain threshold (Scotland et al., 2011). Furthermore, in cultured trigeminal nociceptive neurons, exogenous PRL is able to sensitize transient receptor potential vanilloid 1 (TRPV1) channels and consequently a potential pro-nociceptive effect was suggested (Diogenes et al., 2006).

Indeed, several authors propose that an increase of PRL secretion could contribute to certain pain disorders, such as migraines (Silberstein and Merriam, 1993), rheumatoid arthritis (Chikanza et al., 1993), and mastodynia (Theunissen et al., 2005). Rushen et al. (1993) suggested that an increase of prolactin secretion by suckling paradoxically reduces pain sensitivity. We hypothesize that under non-lactating conditions prolactin might enhance sensory perception, while this might not happen during lactation, due to suckling-induced transformation of the prolactin being secreted from the anterior pituitary. Indeed PRL secreted from the adenopituitary (AP) is transformed into several molecular variants by suckling stimulation (Grosvenor and Mena, 1992). Furthermore PRL has as a number of molecular isoforms produced by posttranslational modifications (Sinha, 1992), and we could propose that this molecular heterogeneity is one of the mechanisms involved in the pleiotropic activity of these peptides in addition to that suggested by Grattan and Kokay (2008) who in part attributed this pleiotropic activity to the regulation of expression of PRL receptors.

Prolactin variants are secreted under different physiological conditions (Mena et al., 2010), and it is known that functional interactions and cytological differences exist among pituitary lactotrophs within the anterior pituitary gland (Dymshitz and Ben-Jonathan, 1991). For instance, molecular isoforms present in the AP of the lactating rats are transformed by suckling stimulation (Mena et al., 1992). This transformation of PRL involves changes in the solubility of the protein that

*Corresponding author. Tel/fax: +52-442-238-1042.
E-mail address: condes@unam.mx (M. Condés-Lara).
URL: http://132.248.142.23/web_site/home_pages/35?locale=en (M. Condés-Lara).
† In memory of Professor Flavio Mena.
Abbreviations: AP, adenopituitary; ELISA, enzyme-linked immunosorbant assay; NS, non-suckled; NS-PRL, prolactin variant from non-sucked rats; NR, non-denaturing; PRL, prolactin; RF, receptive field; S, suckled; SDS–PAGE, sodium dodecyl sulfate–polyacrylamide gel electrophoresis; S-PRL, prolactin variant from suckled rats; TBS, Tris-buffered saline; TRPV1, transient receptor potential vanilloid 1; TTBS, TBS with 0.5% Tween 20; WDR, wide dynamic range neurons.

0306-4522/13 $36.00
http://dx.doi.org/10.1016/j.neuroscience.2013.06.052

confers a longer half-life in the circulation and potentially different biological actions (Mena et al., 1992).

In order to test the potential effect of PRL from the AP of suckled (S) or non-suckled (NS) rats on the neuronal activity of nociceptive fibers, we recorded in anaesthetized male rats the nociceptive responses evoked on the wide dynamic range (WDR) neurons. The WDR neurons are large, second-order neurons that are widely distributed in the dorsal horn spinal cord and receive input from the first-order non-nociceptive (Ab-fibers) and nociceptive (Ad- and C-fibers) neurons (Mendell, 1966; Willis, 1987, 1988). These cells represent an important connector between primary afferent nociceptive fibers and higher nociceptive centers. Our results show that spinal administration of NS-PRL or S-PRL had no effect on the neuronal activity corresponding to the activation of non-nociceptive Ab-fibers (latency between 0 and 20 ms). However, the firing of Ad-fibers (latency between 20 and 90 ms) and C-fibers (90–350 ms) (both nocicceptive) were: (i) increased by NS-PRL and (ii) diminished by S-PRL. Finally, either NS-PRL or S-PRL enhanced the post-discharge activity (350–800 ms). These findings provide a basis for addressing the physiological relevance of the prolactin variants induced by suckling and might help to explain the differences observed in several experiments related to nociception.

EXPERIMENTAL PROCEDURES

Animals

Experiments were carried out on Wistar rats from the Neurobiology Institute Animal House. The animals were housed individually in plastic cages in a special, temperature-controlled room (22 ± 2 °C) and given *ad libitum* access to food (Purina Chow, Ralston Purina Co., Chicago, IL, USA) and water. In the case of primiparous lactating rats, the animals were in a room with a reversed light–dark cycle (lights on at 6:00 h; 14 h light, 10 h darkness). The primiparous rats weighed 250 g at mating, they gained weight during pregnancy and lactation, and they weighed an average of 300 g during experimentation, as did the male rats used in the electrophysiological experiments.

All experimental procedures were approved by our Institutional Ethics Committee, and they were in accordance with the IASP ethical guidelines (Zimmermann, 1983) and the guidelines contained in the NIH guide for the Care and Use of Laboratory animals (80–23, revised in 1996). All efforts were made to limit distress and to use only the number of animals necessary to produce reliable scientific data.

General methods for prolactin extraction

Surgical procedures and preparation of concentrated conditioned media. On postpartum days (10–12), the pups (8–10 pups per litter) from primiparous lactating rats were removed from groups of mothers at 7:00 h, and 6 h later their pups were or were not returned for suckling for 15 min. At the end of the non-suckling (NS) or suckling (S) periods, the mothers were killed by decapitation after light ether anesthesia. The AP was collected using a fine forceps as originally described by Boockfor and Frawley (1987), i.e. the central region around the neurointermediate lobe was dissected and incubated. The AP fragments corresponding to the central pituitary region from S and NS lactating rats were used to prepare the conditioned media. These fragments were incubated in individual flasks containing 300 µl of Earle's medium. Flasks containing the pituitary fragments were gassed with 95% O_2, 5% CO_2, sealed with rubber stoppers, and incubated at 37 °C for 1 h in a water bath shaker (American Optical, Buffalo, NY, USA). After incubation, conditioned media were concentrated, desalted in a Centricon micro-concentrator (Centripep, Millipore, Bredford, MA, USA), and stored frozen until assayed. The PRL concentration in the conditioned media was determined by the enzyme-linked immunoabsorbant assay (ELISA) method (Section 'ELISA').

Sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS–PAGE). The samples of conditioned media from N and NS rats were analyzed by SDS–PAGE in 1.0-mm thick, 6-cm long, 12.5% gels using the buffer system of Laemmli (1970) and Bradford (1976) in a mini Protean III cell (Bio-Rad, California, USA). Samples were electrophoresed under non-denaturing (NR) conditions. The gels were then cut and divided into six fractions (Fig. 1); the proteins in each fraction were electrophoretically eluted, dialyzed, lyophilized, reconstituted, and washed (ProteoSpin, Detergent Clean Up Micro Kit, Norgen, ON, Canada), and assayed by ELISA for PRL content (Fig. 1).

Each fraction of NS-PRL and S-PRL was adjusted to have 20 µg of protein per 100 µl of deionized water, and 20 µl of each solution was administrated at the same spinal cord level (L4–L5) as the electrophysiological recording (see Section 'Western blotting').

ELISA. The PRL concentrations in conditioned media and fractions were determined by the ELISA method as modified by Signorella and Hymer (1984). Briefly, 96-well microtiter plates (Immulon 2HB, Chantilly, VA, USA) were coated overnight at 4 °C with 10 ng of rat PRL in 100 µl of 1 M carbonate buffer, pH 10.3. The plates were washed with TPBS (0.01 M sodium phosphate, 0.15 mM NaCl, 0.05% v/v Tween-20, pH 7). This washing procedure was performed after each incubation step. For the standard curve, serial dilutions of rat PRL (NHPP-NIH) (0.06–64 ng/ml) in TPBS were incubated for 16 h with 100 µl primary anti-rPRL polyclonal antiserum (1:40,000; NHPP-NIH) in TPBS containing 1% (w/v) non-fat dry milk (Bio-Rad). Samples and standards (100 µl) were then added to the coated wells and incubated for 2 h at room temperature. Secondary goat anti-rabbit IgG peroxidase conjugate (Bio-Rad) was then added (1:3000 in TPBS with 1% non-fat dry milk) and incubated for 2 h at room temperature. Bound secondary antibodies were detected by reaction with

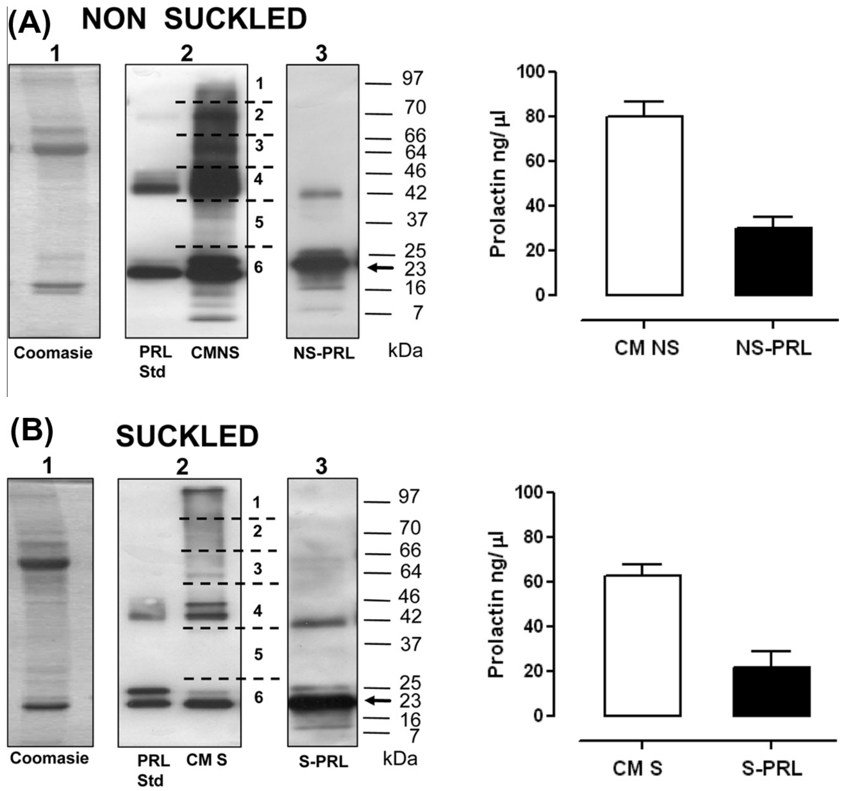

Fig. 1. Representative Western blotting of conditioned medium (CM) of pituitaries from non-suckled (NS) and suckled (S) rats under non-reducing conditions. The stain with Coomassie brilliant blue (A1, B1) shows the main protein bands in the CM. The PRL Std, CM (A2, B2), and electro-eluted gel fraction number 6 (NS-PRL and S-PRL; A3, B3) were examined by western blot using an anti-PRL antibody. Concentrations of PRL in CM and fraction 6 were determined by the ELISA method. No significant differences in PRL concentrations were observed between S and NS samples (n = 5).

2,2′-azino-di-(3-ethylbenzothiazoline sulfonate) substrate (Roche, Mannheim, Germany). Plates were read 15 min later in an automatic ELISA Microplate Reader (Bio-Rad) at 405 nm. The assay has a sensitivity of 2 ng/well and inter-assay and intra-assay coefficients of variation of $<6\%$.

Western blotting. The fractions were electrophoresed under NR conditions by SDS–PAGE. After electrophoresis, the gels were equilibrated in transfer buffer (25 mM Tris–HCl, 192 mM glycine, 20% (v/v) methanol, pH 8.3) for 30–60 min and electrotransferred (at 200 mA for 60 min) to nitrocellulose membranes (Bio-Rad). The membranes were washed with 30 mM Tris, 500 mM NaCl, pH 7.5, (Tris-buffered saline (TBS)) for 15 min, then blocked with 5% (w/v) non-fat dried milk (Bio-Rad) in TBS for 2 h at room temperature. After washing, the membranes were incubated in TBS with 0.5% Tween 20 (TTBS) for 15 min, then overnight at room temperature with anti-rat PRL antiserum, diluted 1:10,000 in TTBS with 1% (w/v) non-fat dry milk. The membranes were then rinsed three times (each 15 min) in TTBS and incubated for 2 h with anti-rabbit IgG–horseradish peroxidase conjugate (Zymed, California, USA) diluted 1:3000 in 1% (w/v) non-fat dry milk in TTBS. Immunoreactive bands were visualized by incubation in ECL chemiluminescent reagent (GE Healthcare, Buckinghamshire, UK) for 5 min and exposed to Kodak Biomax ML film.

General methods for electrophysiological experiments

Surgical procedures. The animals were initially anesthetized in a hermetically sealed box with sevoflurane (3%) in a mixture of two-thirds N_2O and one-third O_2. An intra-tracheal cannula was inserted for artificial ventilation and to maintain the anesthesia throughout the experiment; then, the animals were mounted on a stereotaxic frame. Under these conditions, the animals were secured in a spinal cord unit frame, and lumbar vertebrae were fixed to improve stability at the recording site in order to perform a laminectomy to expose the L_2–L_4 spinal cord segments (Condés-Lara et al., 2003). It is important to mention that the dura was carefully removed. Furthermore, after vehicle or prolactin administration, and in order to avoid desiccation, we covered the exposed spinal cord with mineral oil.

The subsequent experimental protocols (see below) were performed under 2–2.5% sevoflurane to achieve the ethically adequate level of anesthesia without excessively depressing neuronal responses to noxious stimuli (DeLaTorre et al., 2009). The animals were not paralyzed, and we did not observe a withdrawal reaction during the experiments (Condés-Lara et al., 2012). End tidal CO_2 was monitored with a Capstar-100 CO_2 analyzer (CWI Inc., Ardmore, PA, USA) and kept within the physiological range (2.5–3.0%) throughout the experiment. Core body temperature was maintained at 38 °C using a circulating water pad.

Extracellular unitary recordings. Extracellular unitary recordings were made using seven, quartz–platinum/tungsten microelectrodes (impedance 4–7 MΩ) mounted in a multichannel microdrive ''System Eckhorn''. The multielectrode system was manipulated with the 7-channel version of the Fiber-Electrode Manipulator ''System Eckhorn'' using Ekhorn Matrix Multiuser Software (Thomas Recording, Giessen, Germany). The microelectrodes were lowered (200–900 μm from the surface) in small steps (3–6 μm) into the superficial laminae of right dorsal horn segments in search of single-unit discharges.

For each recorded cell, the specific somatic receptive field (RF) was located by tapping on the entire *ipsilateral* hind paw and toes; then, electrical stimulation was applied by two electrodes inserted into the RF (see Section 'Experimental protocol for the electrical stimulation of the somatic RF and administration of exogenous prolactin'). The extracellular neuronal activity induced by electrical stimulation of the RF was recorded, amplified, digitalized, and discriminated using CED hardware and Spike 2 software (Cambridge Electronic Design; Version 5.20). Waveforms and recorded spike trains were stored on a hard drive for off-line analysis.

Experimental protocol for the electrical stimulation of the somatic RF and administration of exogenous prolactin

For electrical test stimulation, two fine needles (27 G) attached to a stimulus isolator unit were inserted subcutaneously into the somatic RF of the recorded neurons. The electrical test stimulation consisted of a train of 20 stimuli at 0.5 Hz, with a 1-ms pulse duration at 1.5 times the threshold intensity required to evoke a C-fiber response. This threshold was determined by giving single electrical pulses with progressive intensity increases (0.1–3.3 mA) until a C-fiber response was evoked.

Under these conditions we recorded the electrophysiological responses of 27 spinal neurons in a control situation consisting of four consecutive (repeated each 10 min), stable electrical responses with variation less than 10%. Then, the 27 neurons were divided into three groups, so that the effects of topical spinal administration of saline (physiological saline, $n = 5$; 20 μl) or one of the prolactin fractions (NS-PRL, $n = 14$ and S-PRL, $n = 8$; 4 μg/20 μl) were analyzed on the neuronal responses induced by the electrical stimulation of the RF (as described above) before and immediately after vehicle or prolactin administration, and every 10 min for the next 90 min. In the case of the vehicle and the NS-PRL fraction, at the end of the 90 min we washed the spinal cord and repeated the procedure detailed above.

Data analysis

The PRL concentration was calculated by linear regression, and values of PRL concentration obtained by ELISA were averaged. Our results are presented as mean ± S.E.M. (standard error of the mean). The number of evoked potentials induced by electrical stimulation of the right posterior paw was expressed as percent change from the respective baseline (100%). The baseline value refers to evoked neuronal response before (control) spinal prolactin administration (NS-PRL or S-PRL). The difference in the neuronal activity evoked within one group of animals before and after prolactin administration was compared using a one-way repeated measures analysis of variance followed, if applicable, by the Student–Newman–Keul post-hoc test. Finally, the areas under curve (AUC) between the first and the second administration of vehicle or NS-PRL fraction were compared using the paired Student's *t*-test. The one-way repeated analysis of variance was followed, if applicable, by the Student–Newman–Keul's post-hoc test. The threshold for statistical significance was set at $P < 0.05$.

RESULTS

Prolactin content of electroeluted prolactin isoforms from AP of lactating S and NS rats

The PRL isoforms released from AP of NS and S rats were analyzed by SDS–PAGE and Western blot as previously shown (Mena et al., 2010, 2011). The main PRL band (fraction 6, Fig. 1) was electroeluted and processed by western blot showing the PRL isoforms. Fraction 6 contained bands ranging from 7 to 25 kDa but primarily a band of 23 kDa and approximately 20 ng/ml of PRL protein (Fig. 1). Spinal administration of this fraction from S or NS rats was used to determine whether it affected neuronal evoked activity.

General properties of recorded cells

From 11 male rats, we recorded 27 spinal cord neurons in the dorsal horn responding to an electrical stimulation of the RF located in the ipsilateral hind paw and toes (Fig. 2A). These cells were recorded over a period of at least 2 h. Five cells (from two rats) were used as control, 14 cells (from six rats) were studied after NS-PRL administration, and eight cells (from three rats) were analyzed after S-PRL administration. The recorded cells showed evoked responses corresponding to the WDR neurons (Fig. 2), and this type of cell presented responses to the activation of Ab-, Ad, C-fiber, and post-discharge activity (see below).

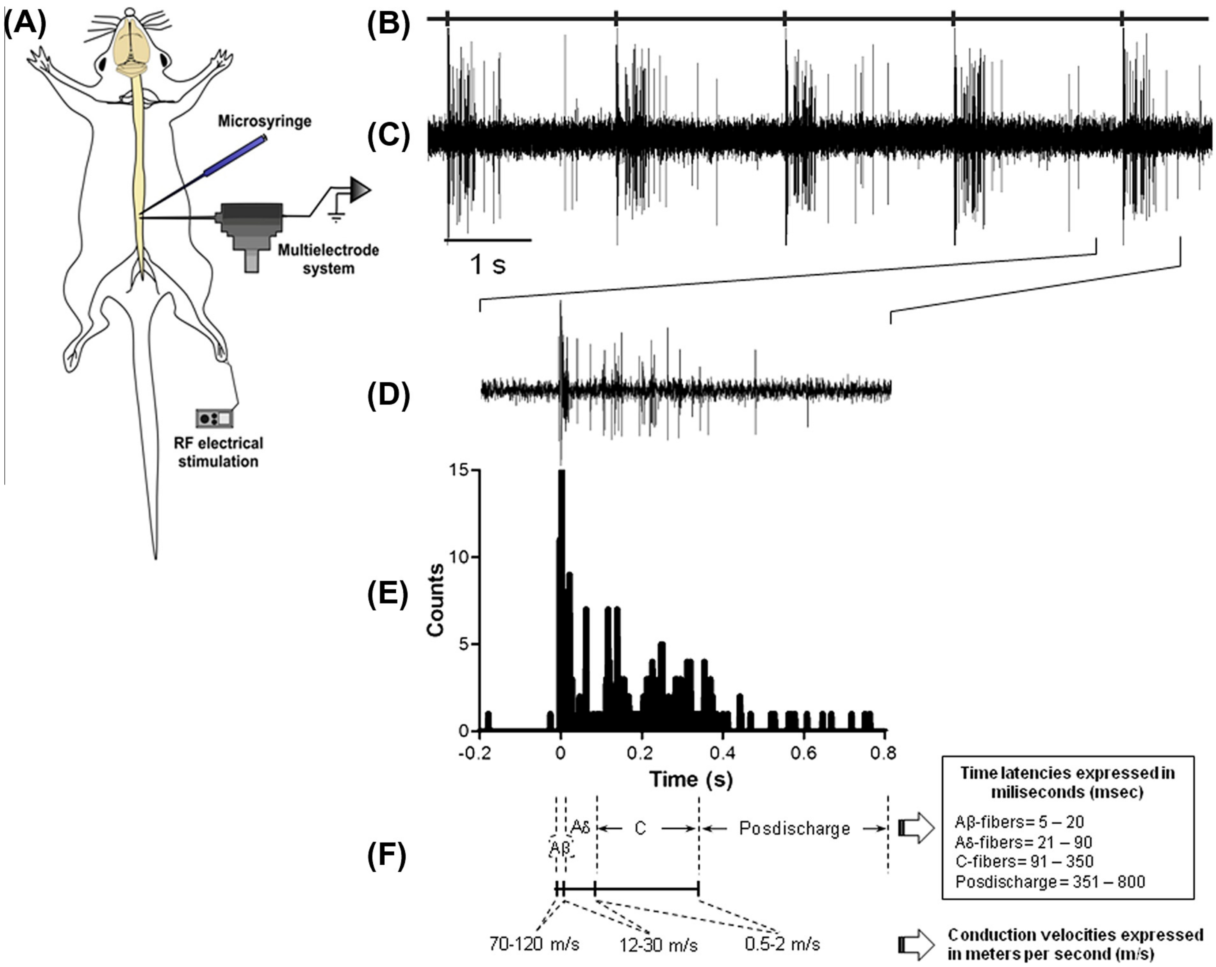

Fig. 2. Neuronal responses to receptive field electrical stimulation (RFst). (A) Experimental set-up showing the neuronal RF stimulation, the recording site in the lumbar spinal cord, and the (microsyringe) injection site; (B) stimulus artifact induced by the electrical stimulation; (C) five consecutive responses to RFst in an identified dorsal horn wide dynamic range (WDR) neuron; (D) a single response to RFst; (E) the peri-stimulus time histogram obtained from the 20 responses to RFst; and (F) time latencies depicting the different fiber components (Ab-fibers, Ad-fibers, C-fibers and post-discharge) of the response induced by the RFst, and the corresponding conduction velocities expressed in m/s.

Neuronal responses evoked by the RF electrical stimulation

Fig. 2 illustrates the neuronal responses to RF electrical stimulation. Fig. 2B shows only five stimulus artifacts, whereas Fig. 2C shows the neuronal tracing obtained (each electrical test stimulation consists of 20 stimuli). In all cases, these neuronal responses were due to selective RF stimulation. The onset of the neuronal responses to electrical stimulation was immediate and consisted of Ab-, Ad, C-fiber, and post-discharge components (Fig. 2C,D). To identify action potentials corresponding to activation of Aβ-fibers (0–20 ms), Aδ-fibers (20–90 ms), C-fibers (90–350 ms), and post-discharge (350–800 ms), the activity of recorded cells was analyzed with peri-stimulus time histograms (1-s duration; bin time = 1 ms, and an offset time of 0.2 s; Spike 2 software) (Fig. 2E). The latencies listed in parentheses are used to identify the fiber type and in Fig. 2 the speed conduction velocities are also shown.

Prolactin effects on the responses evoked by RF electrical stimulation

Figs. 3 and 4 show the analysis of the neuronal responses to RF stimulation before, immediately after, and every 10 min after topical spinal administration of: (i) saline (control; Fig. 3A); (ii) NS-PRL (Fig. 3B), and (iii) S-PRL (Fig. 4). Fig. 3A shows that no time-dependent changes occurred in the neuronal activity in the animals treated with saline. Spinal administration of NS-PRL or S-PRL did not have any effect on the neuronal activity corresponding to Ab-fibers (Figs. 3B and 4, respectively). Interestingly, the Figs. 3B and 5 illustrates that the first application of 4 μg/20 μl NS-PRL produced an enhancement of the neuronal activity of Ad- and C-fibers (P1), which was significantly potentiated after the second administration of NS-PRL (P2). In contrast, Fig. 4 shows that spinal administration of S-PRL significantly inhibited the activity of Ad-fibers after 60 min, whereas the activity of C-fibers was: (i) immediately reduced; (ii) returned to baseline after

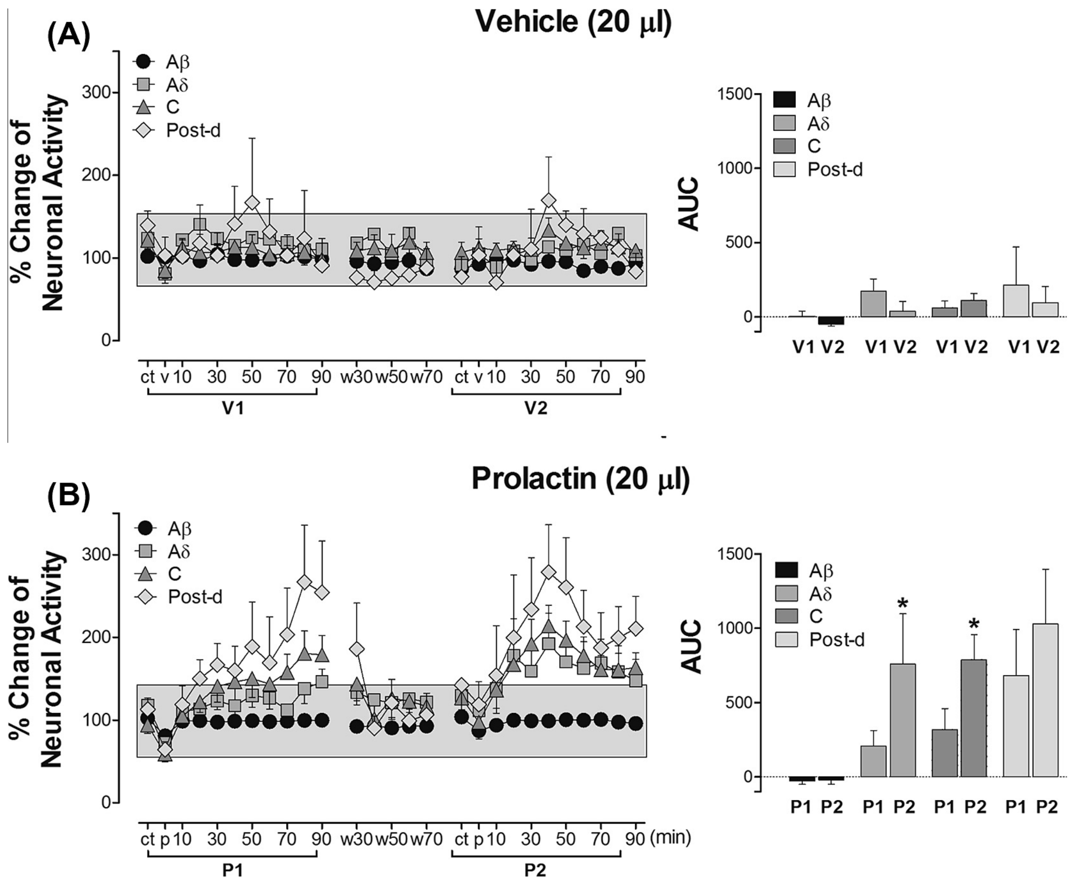

Fig. 3. Effect of administering vehicle (V1 and V2) or NS-PRL (P1 and P2) on the neuronal activity of WDR cells during peripheral receptive electrical field stimulation. (A) The left panel shows the neuronal responses induced electrically before (ct; control), immediately after (v), and every 10 min after the spinal administration of vehicle (20 µl). (B) Left panel shows the neuronal responses induced electrically before (ct; control), immediately after (p), and every 10 min after spinal NS-PRL administration (20 µl). Note that in both treatments, 70 min after wash (w) of the spinal cord, the neuronal activity remains at control levels (ct). Right panels show the area under curve (AUC) of the effects on the neuronal responses induced by the 1st (V1 or P1) and the 2nd (V2 or P2) administration of saline or of NS-PRL. Ab-fibers, Ad-fibers, C-fibers and Post-d post-discharge $^{*}P < 0.05$ vs. control.

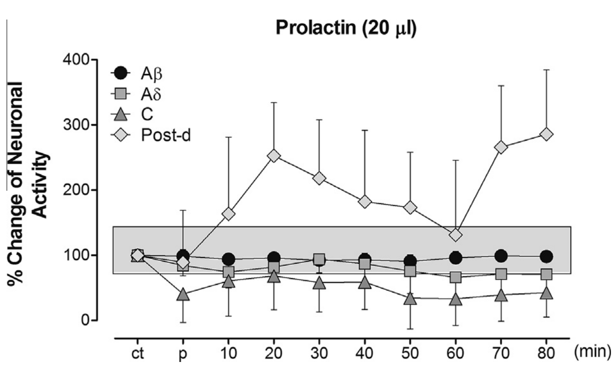

Fig. 4. Percent change of neuronal activity induced by receptive field (RF) electrical stimulation before (control; Ct), immediately after (P), and every 10 min after spinal administration of S-PRL (20 µl). The only significant change was over the post-discharge (Post-d).

10 min; and (iii) inhibited after 60 min. It must be emphasized that after treatments with both NS-PRL and S-PRL, the post-discharge neuronal activity was increased in a time-dependent manner (Figs. 3B and 4).

DISCUSSION

General

Previously, we showed that prolactin isoforms are released into the conditioned media from the central and peripheral AP regions of NS and S lactating rats, and that they selectively stimulate or inhibit the *in vitro* release of PRL by lactotrophs of female or male rats APs (Mena et al., 2010, 2011, 2012), indicating that autocrine effects are exerted by PRL variants contained in electroeluted fractions 1–6 from conditioned medium after SDS–PAGE. In this study we employed the PRL-enriched fraction 6 (Fig. 1), with PRL isoforms ranging from 7 to 25 kDa and other soluble factors, because it had the greatest effects on PRL secretion by APs of rats (Mena et al., 2010). Admittedly, the spinal administration of PRL (∼400 ng) seems like a very high dose. However, one should keep in mind the chemical nature of this molecule and the implications related to: (i) degradation by proteolytic enzymes and (ii) diffusion between the surface of the spinal cord and its site of action.

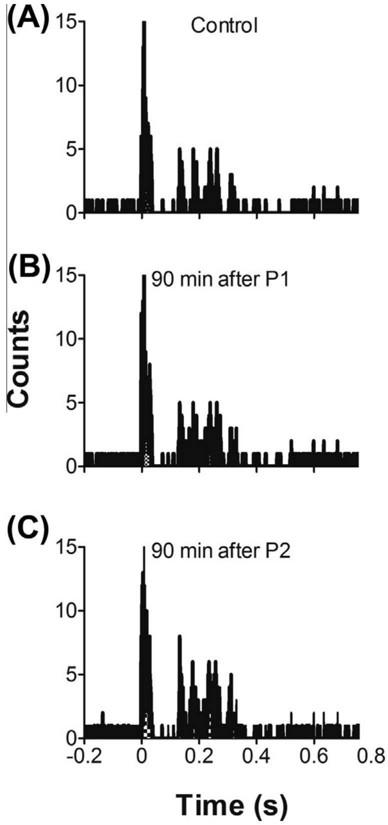

Fig. 5. Firing rate histograms of the same WDR neuron recorded before (A) and after (B, C) spinal administration of NS-PRL. Note that 90 min after the first (P1) spinal administration of NS-PRL, the neuronal activity induced by peripheral electrical stimulation of the receptive field (RF) is enhanced; this increase in the neuronal response is potentiated by a second (P2) administration of NS-PRL.

Apart from the implications discussed below, this study points out the relevance of the physiological condition to obtain reliable results in experiments *in vivo* and sheds further light on the potential reasons to observe a pro-nociceptive (Cruz et al., 1996; Diogenes et al., 2006) or anti-nociceptive (Rushen et al., 1993) effect of prolactin. Additionally, to the best of our knowledge, this study is the first to report that the same fraction of prolactin (coming from S-PRL or NS-PRL) can elicit two different responses on nociceptive neurons.

Effects of the RF electrical stimulation on the neuronal activity of the WDR neurons

Regarding our experimental protocols, we do not measure directly the activity of non-nociceptive Ab-fibers and nociceptive Ad- and C-fibers; instead, we analyze the neuronal activity of the WDR neurons. This type of neuron is located in the dorsal horn of the spinal cord and receives input from the first-order neurons (Ab-, Ad- and C-fibers) (Willis, 1988). This characteristic allows us to establish the differential participation of each type of fiber, depending on the neuronal responses evoked (see Fig. 2).

Under our experimental conditions we observe evoked responses following the electrical RF stimulation (Fig. 2A–C). As previously described (Condés-Lara et al., 2005), in all cases, the onset of the neuronal responses corresponds to Ab-, Ad-, C-fiber, and post-discharge components (Fig. 2D–F). Moreover, these neuronal responses to electrical stimulation were highly reproducible, as they remained essentially without any change over time (Fig. 3A). Thus, no time-dependent changes occurred in the neuronal activity during the course of our experimental protocols. It is interesting that until the present, no description about PRL receptors in the spinal cord is available; to point out at what pre or post-synaptic level PRL is acting. However, the PRL actions seem to be over the nociceptive Ad-, C-fiber and it seems that no action is occurring at the Ab -fiber level in accordance with our electrophysiological recordings. At this point we believe that the PRL actions are presynaptic due to the fact that the recorded neurons continued to respond to Ab- fibers or, alternatively, we need to postulate a differential threshold for the incoming information carried by Ab-, Ad-, and C-fiber. Intra-cellular recordings and PRL receptor study seem to be necessary to understand part of the involved mechanism.

Differential effects of S-PRL and NS-PRL on the nociceptive neuronal evoked responses

The fact that only Ad- and C-fibers, but not Ab-fibers, were modified by the spinal administration of NS-PRL (Fig. 3B) or S-PRL (Fig. 4) suggests that this response: (i) is specific to nociceptive fibers, and (ii) involves a spinal mechanism probably related to a modulator effect on these primary afferent fibers (PAF). In addition, this result raises the question, how does the PRL fraction, purified in the same way from S and NS rats, have different neuronal effects?

Our results show that NS-PRL enhances (Figs. 3B and 5) and the S-PRL inhibits (Fig. 4) the neuronal activity of Ad-, and C-fibers, whereas in both cases, the post-discharge was enhanced. In the first instance, it could be argued that this difference was due to different concentrations of PRL administered, but this seems unlikely since the concentration of PRL in the purified fraction (fraction 6) of NS and S rats is not statistically different (Fig. 1). Previously, it was suggested that an increase of prolactin secretion during lactation reduced pain sensitivity (Rushen et al., 1993); in contrast, during peripheral inflammation it has been reported that PRL can facilitate pain transmission (Scotland et al., 2011). Accordingly, our results support the notion that the paradoxical effect could be attributed to the physiological state of the animal, but the question about how the same fraction has different effects remains open.

Although there is no difference in the PRL concentration and the main protein band in both S and NS rats corresponds to the 23 kDa PRL isoform, we cannot ignore the fact that the same fraction of PRL (from S or NS lactating rats) tested is composed of several variants (see Fig. 1). Indeed, several PRL variants have been characterized and seem to explain, in part, the pleiotropic activity of this neuropeptide (Freeman et al., 2000).

In this respect, our study provides evidence that in S rats a PRL fragment corresponding to line at 16 kDa appears (Fig. 1). At the peripheral level, this variant is able to down-regulate inducible nitric oxide (iNOs) (Lee et al., 2005), an enzyme currently associated with the generation of pain and hyperexcitability of nociceptive fibers (Schmidtko et al., 2009; Hervera et al., 2012). Thus, our results with S-PRL are consistent not only with the study of Rushen et al. (1993), but also with the previous finding showing that the inhibition of nitric oxide (NO) synthesis in the spinal cord led to a reduction of the nociception (Meller and Gebhart, 1993; Luo and Cizkova, 2000).

In contrast, our results for the neuronal effects induced by NS-PRL agree with the behavioral experiments showing that 23 kDa generated during inflammation contributes to hyperalgesia (Scotland et al., 2011), a phenomenon linked to the hyperactivity of nociceptive Ad- and C-fibers (Ji and Woolf, 2001). In addition, as shown in Fig. 3B, a second administration of NS-PRL enhances this neuronal hyperactivity. These findings are consistent with the report showing that repeated exposure of trigeminal nociceptive neurons to PRL results in the amplification (sensitization) of the TRPV1 responses (Diogenes et al., 2006). Indeed, although most of what is known about PRL signaling derives from studies of the mammary gland and several peripheral organs (Bole-Feysot et al., 1998; Brooks, 2012; Ignacak et al., 2012), at neuronal level the PRL-induced depolarization is partially mediated by the: (i) JAK–STAT (Ma et al., 2005); and (ii) PI3K pathways (Lyons et al., 2012). These are signal transduction systems usually associated with the development of neuropathic pain, mechanical allodynia (Dominguez et al., 2008; Yamazaki et al., 2012), and pain behavior (Xu et al., 2007; Pezet et al., 2008; Choi et al., 2010, 2012). Therefore, the increase of neuronal firing of Ad- and C-fibers due to NS-PRL appears plausible.

Additional findings in support of the dual effect of PRL

The heterogeneity in PRL isoforms is physiologically very important in the context of autoregulatory mechanisms that govern the wide range of PRL effects on various nervous structures under different physiological conditions. (Sinha, 1995; Denef, 2008). For example, we reported recently that stimulatory effects upon electrical activity of neurons from the hippocampal and medial preoptic areas were provoked by conditioned medium from APs of S and NS lactating rats but not from male rat APs (Mena et al., 2013). Furthermore, PRL recruits different combinations of electrical and transcriptional responses in neurons, depending upon their anatomical location and phenotype. This may be critical in establishing appropriate responses to PRL under different physiological conditions (Brown et al., 2012).

In Fig. 1 the western blot for fraction 6 of PRL obtained from NS rats shows some other PRL-immunoreactive, but otherwise unidentified compounds that are not observed or have a different chemical conformation in the corresponding fraction from S rats; these small peptides need to be identified in order to determine if and how they participate. At this point the discussion will be very speculative regarding the conformational PRL structural changes as a result of the physiological state of S and NS. There are some possibilities and we only want to make a short list; the enhancement or modification of the molecular activity, possible different transduction mechanisms involved, and participation of different receptors or receptors affinities involved. Further studies are required to address the functional relevance of the same fractions of PRL obtained under different physiological conditions, and this remains an important challenge.

Final considerations about the use of male rats to evaluate the PRL secretion by primaparous lactating rats

It is important to mention that we use male rats to avoid the interference of sex hormones in pain processing (Craft et al., 2004; Vincent and Tracey, 2010; Gupta et al., 2011). Sex hormones are able to alter synaptic transmission, synaptogenesis and receptor function (Aloisi and Bonifazi, 2006; LaCroix-Fralish and Mogil, 2009). Certainly, the use of females to evaluate the effect of both variants of PRL would be relevant, but would require a very large number of animals to cover the different cycling hormonal variations; furthermore, substantial experimental data indicate that female rats usually have a lower pain threshold and pain tolerance (Berkley, 1997; Fillingim et al., 2009; Greenspan et al., 2007), an effect associated with low estrogen levels (LeResche, 2000). In addition, Butcher et al. (1974) showed that PRL secretion changes throughout the 4-day estrous cycle in the rat.

CONCLUSION

Our present results show that PRL can either facilitate or depress the somatic responses of spinal dorsal horn cells, depending on and highlighting the importance of the physiological state of the animal from which it was isolated. Moreover, the PRL effect seems to be selective for the Ad- and C-fibers and also for the post-discharge response evoked by electric stimulation in the RF of the recorded neurons.

CONFLICT OF INTEREST

The authors state that they have no conflict of interest.

Acknowledgements—We gratefully acknowledge Dr. Dorothy D. Pless for revising grammatical aspects of the manuscript and Biol. Eugenia Ramos for technical assistance. We are also indebted to CONACyT for Grants 128037 (F.M.) and 164536 (M.C-L.) and PAPIIT-DGAPA UNAM Grant IN206711 (F.M.). A.G-H. is a Postdoctoral Fellow of DGAPA-UNAM.

REFERENCES

Aloisi AM, Bonifazi M (2006) Sex hormones, central nervous system and pain. Horm Behav 50:1–7.

Ben-Jonathan N, Mershon JL, Allen DL, Steinmetz RW (1996) Extrapituitary prolactin: distribution, regulation, functions, and clinical aspects. Endocr Rev 17:639–669.

Berkley KJ (1997) Sex differences in pain. Behav Brain Sci 20:371–380.

Bole-Feysot C, Goffin V, Edery M, Binart N, Kelly PA (1998) Prolactin (PRL) and its receptor: actions, signal transduction pathways and phenotypes observed in PRL receptor knockout mice. Endocr Rev 19:225–268.

Boockfor FR, Frawley LS (1987) Functional variations among prolactin cell from different pituitary regions. Endocrinology 120:874–879.

Bradford MM (1976) A rapid and sensitive method for the quantitation of micrograms quantities of protein utilizing the principle of protein –dye binding. Anal Biochem 7(248):254.

Brooks CL (2012) Molecular mechanisms of prolactine and its receptors. Endocrine Rev 33:504–525.

Brown RS, Piet R, Herbison AE, Grattan DR (2012) Differential actions of prolactin on electrical activity and intracellular signal transduction in hypothalamic neurons. Endocrinology 153:2375–2384.

Butcher RL, Collins WE, Fugo NW (1974) Plasma concentration of LH, FSH, prolactin, progesterone and estradiol-17b throughout the 4-day estrous cycle of the rat. Endocrinology 94:1704–1708.

Chikanza IC, Petrou P, Chrousos G, Kingsley G, Panayi GS (1993) Excessive and dysregulated secretion of prolactin in rheumatoid arthritis: immunopathogenetic and therapeutic implications. Br J Rheumatol 32:445–448.

Choi JI, Svensson CI, Koehrn FJ, Bhuskute A, Sorkin LS (2010) Peripheral inflammation induces tumor necrosis factor dependent AMPA receptor trafficking and Akt phosphorylation in spinal cord in addition to pain behavior. Pain 149:243–245.

Choi JI, Koehrn FJ, Sorkin LS (2012) Carrageenan induced phosphorylation of Akt is dependent on neurokinin-1 expressing neurons in the superficial dorsal horn. Mol Pain 8:4.

Condés-Lara M, Marina-González N, Martínez-Lorenzana G, Luis-Delgado O, Freund-Mercier MJ (2003) Actions of oxytocin and interactions with glutamate on spontaneous and evoked dorsal spinal cord neuronal activities. Brain Res 976:75–81.

Condés-Lara M, Maie IA, Dickenson AH (2005) Oxytocin actions on afferent evoked spinal cord neuronal activities in neuropathic but not in normal rats. Brain Res 1045:124–133.

Condés-Lara M, Rojas-Piloni G, Martínez-Lorenzana G, Diez-Martínez DC, Rodríguez-Jiménez J (2012) Functional interactions between the paraventricular hypothalamic nucleus and raphe magnus. A comparative study of an integrated homeostatic analgesic mechanism. Neuroscience 209:196–207.

Craft RM, Mogil JS, Aloisi AM (2004) Sex differences in pain and analgesia: the role of gonadal hormones. Eur J Pain 8:397–411.

Cruz Y, Martínez-Gómez M, Manzo J, Hudson R, Pacheco P (1996) Change in pain threshold during the reproductive cycle of the female rat. Physiol Behav 59:543–547.

DeLaTorre S, Rojas-Piloni G, Martínez-Lorenzana G, Rodríguez-Jiménez J, Villanueva L, Condés-Lara M (2009) Paraventricular oxytocinergic hypothalamic prevention or interruption of long-term potentiation in dorsal horn nociceptive neurons: electrophysiological and behavioral evidence. Pain 144:320–328.

Denef C (2008) Paracrinicity: the story of 30 years of cellular pituitary crosstalk. J Neuroendocrinol 20:1–70.

Diogenes A, Patwardhan AM, Jeske NA, Ruparel NB, Goffin V, Akopian AN, Hargreaves KM (2006) Prolactin modulates TRPV1 in female rat trigeminal sensory neurons. J Neurosci 26:8126–8136.

Dominguez E, Rivat C, Pommier B, Mauborgne A, Pohl M (2008) JAK/STAT3 pathway is activated in spinal cord microglia after peripheral nerve injury and contributes to neuropathic pain development in rat. J Neurochem 107:50–60.

Dymshitz J, Ben-Jonathan N (1991) Effects of coculture of anterior and posterior pituitary cells on the responsiveness of lactotrophs to different secretagogues. Endocrinology 129:2340–2535.

Fillingim RB, King CD, Ribeiro-Dasilva MC, Rahim-Williams B, Riley 3rd JL (2009) Sex, gender, and pain: a review of recent clinical and experimental findings. J Pain 10:447–485.

Freeman ME, Kanyicska B, Lerant A, Nagy G (2000) Prolactin: structure, function, and regulation of secretion. Physiol Rev 80:1523–1631.

Grattan DR, Kokay IC (2008) Prolactin: a pleiotropic neuroendocrine hormone. J Neuroendocrinol 20:752–763.

Greenspan JD, Craft RM, LeResche L, Arendt-Nielsen L, Berkley KJ, Fillingim RB, Gold MS, Holdcroft A, Lautenbacher S, Mayer EA (2007) Studying sex and gender differences in pain and analgesia: a consensus report. Pain 132(Suppl. 1):S26–S45.

Grosvenor CE, Mena E (1992) Regulation of prolactin transformation in the rat pituitary. In: Functional anatomy of the neuroendocrine hypothalamus. England: John Wiley & Sons Ltd.. p. 69–80 [Ciba Found Symp 168].

Gupta S, McCarson KE, Welch KM, Berman NE (2011) Mechanism of pain modularion by sex hormones in migraine. Headache 51:869–879.

Hervera A, Leánez S, Pol O (2012) The inhibition of the nitric oxide–cGMP–PKG–JNK signaling pathway avoids the development of tolerance to the local antiallodynic effects produced by morphine. Eur J Pharmacol 685:42–51.

Ignacak A, Kasztelnik M, Sliwa T, Korbut RA, Rajda K, Guzik TJ (2012) Prolactin – not only lactotrophin. A ''new'' view of the ''old'' hormone. J Physiol Pharmacol 63:435–443.

Ji RR, Woolf CJ (2001) Neuronal plasticity and signal transduction in nociceptive neurons: implications for the initiation and maintenance of pathological pain. Neurobiol Dis 8:1–10.

Lacroix-Fralish ML, Mogil JS (2009) Progress in genetic studies of pain and analgesia. Annu Rev Pharmacol Toxicol 49:97–121.

Laemmli UK (1970) Cleavage of structural proteins during the assembly of the head of bacteriophage T4. Nature 227:680–685.

Lee S-H, Nishino M, Mazumdar T, García GE, Galfione M, Lee FL, Lee C, Liang A, Kim J, Feng L, Eissa NT, Lin S-H, Yu-Lee L-Y (2005) 16-kDa prolactin down-regulates inducible nitric oxide expression through inhibition of the signal transducer and activator of transcription 1/IFN regulatory factor-1 pathway. Cancer Res 65:7984–7992.

LeResche L (2000) Epidemiologic perspectives on sex differences in pain. In: Fillingim RB, editor. Sex, gender and pain, progress in pain research and management, vol. 17. Seattle: IASP Press. p. 233–249.

Luo ZD, Cizkova D (2000) The role of nitric oxide in nociception. Curr Rev Pain 4:459–466.

Lyons DJ, Hellysaz A, Broberger C (2012) Prolactin regulates tuberoinfundibular dopamine neuron discharge pattern: novel feedback control mechanisms in the lactotrophic axis. J Neurosci 32:8074–8083.

Ma FY, Anderson GM, Gunn TD, Goffin V, Grattan DR, Bunn SJ (2005) Prolactin specifically activates signal transducer and activator of transcription 5b in neuroendocrine dopaminergic neurons. Endocrinology 146:5112–5119.

Meller ST, Gebhart GF (1993) Nitric oxide (NO) and nociceptive processing in the spinal cord. Pain 52:127–136.

Mena F, Hummelt G, Aguayo D, Clapp C, Martínez de la Escalera G, Morales MT (1992) Changes in molecular variants during in vitro transformation and release of prolactin by the pituitary gland of the lactating rat. Endocrinology 130:3365–3377.

Mena F, Navarro N, Castilla A, Morales T, Fiordelisio T, Cárabez A, Aguilar MB, Huerta-Ocampo I (2010) Prolactin released in vitro from the pituitary of lactating, pregnant, and steroid-treated female or male rats stimulates prolactin secretion from pituitary lactotropes of male rats. Neuroendocrinology 91:77–93.

Mena F, Navarro N, Castilla A (2011) Regulatory effects of dopamine, oxytocin and thyrotropin-releasing hormone on the release of prolactin variants from the adenohypophysis of lactating rats. Int J Endocrinol Metab 9:382–390.

Mena F, Navarro N, Castilla A (2012) Autocrine regulation of prolactin secretion by prolactin variants released from lactating rat adenohypophysis. Acta Endocrinol 8:199–214.
Mena F, Navarro N, Castilla A (2013) Autocrine and paracrine regulation of prolactin secretion by prolactin variants and by hypothalamic hormones. In: György MN, Bela ET, editors. Prolactin. London: InTech. p. 97–120.
Mendell LM (1966) Physiological properties of unmyelinated fiber projection to the spinal cord. Exp Neurol 16:316–332.
Pezet S, Marchand F, D'Mello R, Grist J, Clark AK, Malcangio M, Dickenson AH, Williams RJ, McMahon SB (2008) Phosphatidylinositol 3-kinase is a key mediator of central sensitization in painful inflammatory conditions. J Neurosci 28:4261–4270.
Rushen J, Foxcroft G, De Passillé AM (1993) Nursing-induced change in pain sensitivity, prolactin, and somatotropin in the pig. Physiol Behav 53:265–270.
Schmidtko A, Tegeder I, Geisslenger G (2009) No NO, no pain? The role of nitric oxide and cGMP in spinal pain processing. Trends Neurosci 32:339–346.
Scotland PE, Patil M, Belugin S, Henry MA, Goffin V, Hargreaves KM, Akopian AN (2011) Endogenous prolactin generated during peripheral inflammation contributes to thermal hyperalgesia. Eur J Neusrosci 34:745–754.
Signorella AP, Hymer WC (1984) An enzyme-linked immunosorbent assay for rat prolactin. Anal Biochem 136:372–381.
Silberstein SD, Merriam GR (1993) Sex hormones and headache. J Pain Symptom Manage 8:98–114.
Sinha YN (1992) Prolactin variants. Trends Endocrinol Metab 3:100–106.
Sinha YN (1995) Structural variants of prolactin: occurrence and physiological significance. Endocr Rev 16:354–369.
Theunissen C, De Schepper J, Schiettecatte J, Verdood P, Hooghe-Peeters EL, Velkeniers B (2005) Macroprolactinemia: clinical significance and characterization of the condition. Acta Clin Belg 60:190–197.
Vincent K, Tracey I (2010) Sex hormones and pain: the evidence from functional imaging. Curr Pain Headache Rep 14:396–403.
Willis WD (1987) The spinothalamic tract in primates. In: Besson J-M, Guilbaud G, Peschanski M, editors. Thalamus and pain. Amsterdam: Elsevier. p. 35–57.
Willis WD (1988) Anatomy and physiology of descending control of nociceptive responses of dorsal horn neurons: comprehensive review. In: Fields HL, Besson J-M, editors. Pain modulation, progress in brain research, vol. 77. Amsterdam: Elsevier. p. 1–29.
Xu JT, Tu HY, Xin WJ, Liu XG, Zhang GH, Zhai CH (2007) Activation of phosphatidylinositol 3-kinase and protein kinase B/Akt in dorsal root ganglia and spinal cord contributes to the neuropathic pain induced by spinal nerve ligation in rats. Exp Neurol 206:227–269.
Yamazaki S, Yamaji T, Murai N, Yamamoto H, Matsuda T, Price RD, Matsuoka N (2012) FK1706, a novel non-immunosuppressive immunophilin ligand, modifies gene expression in the dorsal root ganglia during painful diabetic neuropathy. Neurol Res 34:469–477.
Zimmermann M (1983) Ethical guidelines for investigation of experimental pain in conscious animals. Pain 16:109–110.

(Accepted 23 June 2013)
(Available online 02 July 2013)